

Gianmarco Santoro

AI Engineer focused on **Computer Vision**, **Robotics**, and applied **ML**

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Applied AI Advisor

Startup (Confidential) | 2025 – Present

- Supporting an early-stage startup which helps companies **adopting AI to automate workflows** and improve processes. Case study:
 - Rapid prototyping of lightweight **proof-of-concept** system to **automate CNC programming** workflows. Proposed an architecture to extract features from CADs, G-codes and technical sheets to **fine-tune a compact LLM** (e.g.T5-small) for task-specific code generation. Analysed multiple solutions including GNN/transformer models, and an alternative computer-vision approach (VLMs).

DeepFake Detection | Computer Vision - Machine Learning Engineer

MICC | Florence, Italy | 2023 – 2024 | Python, PyTorch, OpenCV, NumPy, Pandas, Matplotlib, CUDA, GitHub, SSH

- Implemented **Out-of-Distribution (OOD) Detection** on *FaceForensics++* dataset to identify AI-generated images. A 3-step pipeline:
 - Extracted Self-Attention heatmaps via **Vision Transformer**, pre-trained on **face recognition**.
 - Exploited heatmaps: trained a **Convolutional Autoencoder** on real ones.
 - OODD: tested real vs fake reconstruction error, achieved **60% AUROC across 5 forgeries**, showing transfer learning capabilities.

Data Science & Statistical Learning | 2nd Level Master - Postgraduate

UniFi & IMT for Advanced Studies | 2023 - 2024 | 110/110 cum laude | Python, PyTorch, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Jupyter/Colab, R & RStudio, G-Suite

- Some topics: **Optimization**, GIS, NLP, Graph Theory. • Projects (visit [Website](#) for more):
 - Developed **Deep Learning images classifier** via **CNNs** models (PyTorch), achieving an accuracy of 70.4% on the test set.
 - Built **ML housing price pipeline**: K-Fold Cross-Validation, **Grid Search** on polynomial regression degree improving MAE by 12%.
 - Predicted salary** on selected features using **CART** and other models. Reached **83.8% accuracy** with Gradient Boosting.
 - Forecasted **post-booking classes attendance**: applied **logistic regression** on a highly imbalanced dataset.

Smart Center | SW & System Integration Engineer

Daikin Applied Europe SpA | Rome, Italy | 2021 - 2022 | Python, GitLab, PLC Siemens-C., bash, TCP/IP, mRemoteNG, Modbus, qModMaster

- Worked on cloud-based HVAC monitoring, system integration, and services:
 - Automated PLC** mapping and alarm selection via **Python**, reducing manual configuration time by ~95%.
 - Built real-time **IoT sensor-to-cloud temperature control system** integrating 21 Modbus sensors, removing 100% manual setup.
 - Aggregated data via web API to generate 2000+ reports/month, improving issues detection and reducing workload by ~80%.

Wireless Charging System | Operations Research Scientist

Enermove Srl | Turin, Italy | 2020 - 2021 | Matlab, Optimization Toolbox, MS-Office (Excel), AutoCAD

- Optimized electric vehicles wireless charging system topology** via **Genetic Algorithms**, improving energy efficiency-to-cost ratio.

Chip Extrusion Recycling | Manufacturing Research Engineer

Institut für Umformtechnik und Leichtbau | Dortmund, Germany | 2018 | Matlab, Excel, CATIA, HyperMesh/Extrude, Lab instruments

- Experimented with the **aluminum recycling process** by direct extrusion of mixed waste alloys (DoE). Compared measured properties with FEM and numerical results. Assessed **industrial feasibility** identifying process bottlenecks: logistic issues and product defects.

Mechanical Engineering | M.Sc. at PoliTo & B.Sc. at UniBo | 104/110 | Matlab & Simulink, AutoCAD, SolidWorks, MSC-Adams/Apex

- Specialized in **Automation**. Some extra projects:
 - Automation and **mechatronics** labs: PLC, PID controls, pneumatics, mechanisms.
 - Designed a reduction **gearbox**.
 - Developed a home **heating system**.
 - Built a **numerical model** for metal flue heat exchange.

Side Projects | 2022 - Current | Figma, Mural, Notion, Canva, Google-Forms/Sheets/Docs, HTML, CSS, JS, zsh, GIT & GitHub

- Conducted Market Research, Javelin Board and SWOT analysis, Lean Startup Method on PoC/MVPs: **Project-Driven Work Community**; **Local Food** Logistics Platform; **Food-based** Social Media; **Selected Products** Referral Platform.

Languages: • **Italian:** C2 - Mother tongue • **English:** C1 - Professional • **German:** A2 - Basic